



Degradation modeling of serial space lithium-ion battery pack based on online inconsistency representation parameters

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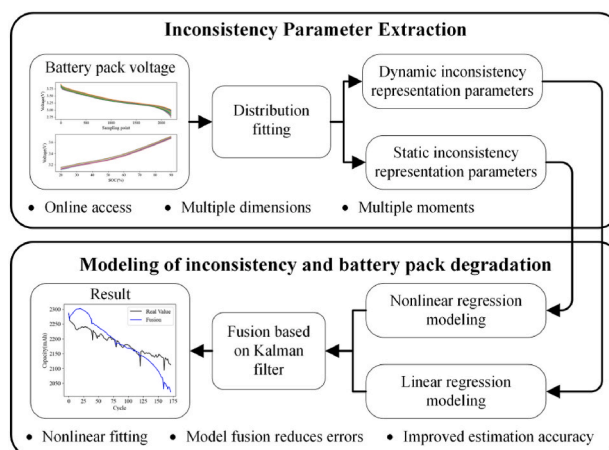
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HIGHLIGHTS

- A method for modeling lithium-ion battery pack degradation based on inconsistency.
- Online representation of static and dynamic inconsistency via voltage.
- A fusion model combining linear and nonlinear approaches enhances model accuracy.
- Cross-validation shows MAE and ME results better than 1.73 % and 3.31 %.

GRAPHICAL ABSTRACT



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ABSTRACT

Establishing an inconsistency-based degradation model for lithium-ion battery packs is crucial for suppressing the degradation of battery packs by optimizing the inconsistency. This paper proposes a method for modeling the degradation of serial space lithium-ion battery packs based on online inconsistency representation parameters. Firstly, the static inconsistency representation parameters are acquired online by quantifying the difference among voltage intervals of cells through Gaussian distribution, addressing the challenge of acquiring static inconsistency representation parameters online. Simultaneously, dynamic inconsistency representation parameters are serialized representations by quantifying the voltage differences of cells at multiple moments, better capturing the rapid evolutionary process of inconsistency. Secondly, a linear model is used to model the serialized parameters and degradation, simplifying the modeling process while achieving multi-source information fitting. Meanwhile, a nonlinear model is constructed to better fit the nonlinear degradation trend of the battery pack. Then, Kalman filter is utilized for model fusion to achieve complementary advantages and improve modeling accuracy. Cross-validation with laboratory battery pack data confirms that this method outperforms

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comparison approaches, achieving the mean absolute error and maximum error of less than 1.73 % and 3.31 %, respectively. The method proposed in this paper provides a basis for future work on battery pack life extension.

1. Introduction

Lithium-ion batteries are widely utilized in space applications, such as satellites and space stations, due to their long cycle life [1,2]. Lithium-ion batteries are not typically used as battery cells. To meet system power requirements, serial lithium-ion battery packs have become a primary configuration in space applications. With the continuous charge and discharge process, the performance of lithium-ion batteries gradually degrades, primarily indicated by the decrease in maximum discharge capacity and the increase in internal resistance [3]. The decrease in maximum discharge capacity is the most common indicator of performance degradation. Therefore, maximum discharge capacity is used in this paper to describe battery performance degradation. Lithium-ion battery pack performance degradation is influenced not only by cell performance degradation but also by inconsistency, which reduces overall performance and accelerates pack degradation [4].

Inconsistency among lithium-ion battery cells is unavoidable due to differences in production processes and operating environments. Inconsistency is primarily reflected in parameters such as internal resistance, capacity, state of charge, etc. Among cells [5,6], Thorsten et al. [7] tested 48 identical battery cells under the same operating conditions. After 1000 cycles, the capacity difference among the battery cells reached 10 %. Moreover, inconsistency among lithium-ion battery cells accelerates the degradation of battery packs. Gogoana et al. [8] experimented with the effects of capacity and internal resistance inconsistency on battery pack cycle life. Experiment results indicated that a 20 % mismatch in internal resistance and capacity among lithium-ion battery cells leads to a 40 % degradation in battery pack life. Thus, capacity and internal resistance have become core parameters in the selection of battery cells for packs. Ganesan et al. [9] examined the effects of temperature differences on battery pack performance. The study results indicated that when the temperature difference among the cells is 15 °C, capacity loss in the battery pack increases by 5 %. In summary, the impact of inconsistency on battery pack performance degradation is significant. Thus, quantifying the relationship between inconsistency and battery pack performance degradation is essential. This work provides a basis for the principles of battery cell selection, estimation of battery pack degradation states, and suppression of battery pack degradation by optimizing inconsistency.

The basis for quantifying the relationship between inconsistency and battery pack degradation is the acquisition of battery pack inconsistency. However, the inconsistency of lithium-ion battery packs cannot be acquired directly and requires indirect representation [10,11]. The acquisition of inconsistency representation parameters needs to be achieved using entropy-based methods, distance-based methods, distribution-based methods, and other methods [12–14]. The inconsistency representation parameters obtained through the above methods are usually categorized into static inconsistency representation parameters and dynamic inconsistency representation parameters [15]. Static inconsistency representation parameters change relatively slowly and take a long time to change significantly. Static inconsistency representation parameters are usually constructed relying on capacity, internal resistance, etc [16]. Dynamic inconsistency representation parameters change relatively quickly and can change significantly in a short period. Dynamic inconsistency representation parameters are usually constructed relying on the state of charge, current moment voltage, current moment current, etc [17]. Wang et al. [18] fitted the voltage distribution at a specific moment for lithium-ion batteries and utilized the distribution parameters as dynamic inconsistency representation parameters. Tian et al. [15] clustered the battery capacity under offline testing to

enable the acquisition of static inconsistency representation parameters. Wang et al. [19] obtained the internal resistances of cells using special excitation and represented the differences among the internal resistances using distances. Then, the acquisition of static inconsistency representation parameters was achieved by the above operation. Hou et al. [20] performed an inconsistency representation of capacity, internal resistance, and constant-current charging time in an offline state. Thus, static inconsistency representation parameters and dynamic inconsistency representation parameters were obtained. Although there is a lot of existing research on inconsistency representations, there is still some room for progress. For example, the dynamic inconsistency representation parameters are only single moment data, which cannot represent the dynamic change process; the static inconsistency representation parameters are constructed for internal resistance, capacity, and other parameters, which are not available in space application scenarios, where specific discharge tests cannot be performed, and therefore the above parameters are not available online.

The relationship between inconsistency and battery pack degradation remains unclear, with most studies providing qualitative analyses. Kwon et al. [21] identified voltage inconsistency using the ICA method and trained an LSTM model sensitive to inconsistency to mitigate its effects. This approach improved the accuracy of battery pack degradation state estimation. Wang et al. [22] developed a mechanism-level mathematical model to describe the dynamic evolution of lithium-ion battery degradation, providing a foundation for analyzing battery degradation. Xu et al. [23] employed wavelet analysis to effectively estimate the degradation of battery pack performance. This method analyzed the differences in degradation among individual cells, thereby enhancing the accuracy of the degradation estimation. Wang et al. [24] analyzed the differences among batteries at various stages and introduced an adaptive optimization mechanism in the degradation modeling to minimize the impact of these differences. This led to the establishment of a high-accuracy and high-stability battery degradation modeling method, which holds significant reference value. Fan et al. [25] utilized randomly generated parameters to model the equivalent circuit of the battery pack. Based on this model, a qualitative analysis of lithium-ion battery pack inconsistency was conducted. Wang et al. [26] examined the evolution patterns of inconsistency in both series and parallel lithium-ion battery packs, providing a foundation for future quantitative investigations into the impact of inconsistency on battery pack degradation. In addition to analyzing inconsistency and performance degradation, researchers have also examined inconsistency relationship with other state parameters, such as SOC and power state [27–30]. Currently, only a limited number of studies have quantitatively analyzed the inconsistency and degradation of battery packs. Wang et al. [31] used linear regression models to analyze the degradation relationship between battery cell capacity and combined battery pack capacity. Che et al. [32] obtained 15 inconsistency parameters and applied Gaussian Process Regression to establish the relationship between these inconsistency parameters and performance state degradation. In summary, most of the existing studies have focused on reducing the effect of inconsistency on state estimation results or qualitatively analyzing the correlation between the two. The relationship between quantitative inconsistency and degradation is insufficiently studied. Moreover, existing quantitative analyses often overlook the nonlinear degradation of lithium-ion battery packs.

Therefore, existing studies have developed a large number of research results in the areas of inconsistency representation and degradation modeling. Nevertheless, there are areas in current research that could be further enhanced, as outlined below.

Table 1
Lithium-ion battery pack test conditions.

Steps	Pack1	Pack2	Pack3
1	0.6C (1.32A) constant current discharge to the minimum voltage of the cell in the battery pack $\leq 2.75\text{V}$		
2	Rest 5 min		
3	0.8C (1.76A) constant current charging to the maximum voltage of the cell in the battery pack $\geq 4.2\text{V}$	1.0C (2.2A) constant current charging to the maximum voltage of the cell in the battery pack $\geq 4.2\text{V}$	
4	Rest 5 min		
5	4.2V constant voltage charging to current $\leq 0.1\text{A}$		
6	Rest 5 min		
7	Cycle step 1 to step 6 for a total of 40 times, then disassemble the battery pack and perform degradation tests on the battery cells		
8	Cycle step 1 to step 7		

- (1) Lithium-ion battery packs lack online representation methods for multiple dimensions and multiple moments of inconsistency. Currently, the dynamic inconsistency representation parameter is limited to a single moment, while dynamic inconsistency evolves rapidly and lacks a serialized representation approach. Static inconsistency representation parameters depend solely on offline testing, which hinders online parameter acquisition.
- (2) Quantitative modeling overlooks the nonlinear characteristics of the battery pack degradation process. Moreover, most existing models of lithium-ion battery pack inconsistency and performance focus on mitigating the effects of inconsistency rather than establishing a coupling between the two.

To address the above problems, this paper constructs a multiple dimensions and multiple moments inconsistency representation parameter cluster, based on which linear and nonlinear models are fused to achieve accurate modeling of the battery pack degradation process. The main contributions of this paper are as follows.

- (1) First, this study indirectly represents the depth of discharge (DOD) of each cell using online monitorable voltage and quantifies the differences in DOD using Gaussian distribution. This enables the online representation of static inconsistency, effectively compensating for the limitations of static parameters like capacity and internal resistance, which cannot be monitored online. Then, the evolution of the voltage difference of each cell is described from multiple moments, thus realizing the serialized representation of the dynamic inconsistency of the battery pack. Through the above operations, this study realizes the online representation of inconsistency from multiple dimensions and multiple moments.
- (2) First, this study models the serialized dynamic inconsistency representation parameters and the battery pack degradation using a linear fitting model, which simplifies the modeling process while achieving the fitting of multi-source information. Then, a nonlinear fitting model is used to model the static inconsistency representation parameters with the battery pack degradation to capture the nonlinear degradation characteristics of the battery pack. Finally, the above two models are fused using the Kalman filter, which realizes the complementary advantages of multi-source information fitting and nonlinear fitting ability, thus improving the model accuracy.

The subsequent sections of the paper are organized as follows: Section 2 provides a detailed description of the battery pack test and data analysis. Section 3 begins by outlining the overall framework of the

proposed modeling approach and subsequently delves into a comprehensive analysis and description of its various aspects. Section 4 details the experimental setup, under which experiments are conducted and results are compared and analyzed. Section 5 offers a summary of the paper and outlines the next phase of the work.

2. Battery pack test and data discussions

Serial lithium-ion battery packs are widely used in space applications. Thus, the experimental subject of this paper is a battery pack constructed by connecting four cells in series. Each lithium-ion cell has a maximum discharge capacity of 2200 mAh and a nominal voltage of 3.6 V. The upper voltage limit is 4.2 V, and the lower voltage limit is 2.75 V. Before assembling the battery pack, the cells must undergo a screening process to minimize inconsistency. A discharge test was conducted on 60 new battery cells to determine their maximum discharge capacity. Twelve cells with the smallest capacity variation were selected from the 60 tested cells [33,34]. Three battery packs were assembled from the 12 selected cells, with four cells connected in series per pack. The charging and discharging conditions for the battery packs in the cycle test are shown in Table 1.

The battery pack charging and discharging equipment used in this study is the CT-4004 from NEWARE Ltd. The battery cell charging and discharging equipment is the CT-3008, also from NEWARE Ltd. The temperature chamber used is the BTH-1000 from Dongguan Bell Ltd. The test equipment is shown in Fig. 1.

As shown in Table 1, three sets of battery pack charge/discharge tests were conducted in this study. Battery packs 2 and 3 operated under the same conditions, while battery pack 1 had a lower charging rate. All three battery packs cycled 172 cycles, with the capacity degradation data shown in Fig. 2(a).

As shown in Fig. 2(a), the differences in battery pack degradation are minimal due to the similar working conditions across the packs. Battery pack 3 is selected for detailed degradation analysis, with its data separately plotted in Fig. 2(b).

(1) Analysis of data fluctuations

Battery pack 3 exhibited significant fluctuations at the 40th, 80th, 120th, and 160th cycles. The reason for this phenomenon was that the battery pack was disassembled and tested for cell degradation state at 40 cycle intervals. The state of each cell needed to be recorded before disassembly. After performing degradation tests on the disassembled cells, they were returned to their original voltage state. As a result, fluctuations in battery degradation data occurred before and after disassembly. Other fluctuations in the data were attributed to equipment maintenance, power outages, and varying hold durations, which led to phenomena such as battery capacity rebound.

(2) Analysis of degradation trends

During the early cycles (1st-17th), the battery's capacity experienced a significant drop. This behavior was characteristic of initial battery aging, likely caused by SEI film formation and the activation of electrode material. During the middle cycles (18th-120th), despite fluctuations, the capacity continued to degrade at a relatively steady rate. In the later cycles (121th-172nd), the degradation of the battery pack significantly slowed due to strong capacity recovery. As cycling continues, the degradation of battery cells was expected to accelerate, increasing cell inconsistency and hastening battery pack degradation.

The battery pack is influenced by environmental factors, cells degradation variations, and differences in operating conditions. There is uncertainty in both the degradation rate and trend of the battery pack. The overall degradation process of the battery pack is nonlinear, and this nonlinear increases as the number of cycles increases. The methodology proposed in this paper is validated using the three data sets mentioned

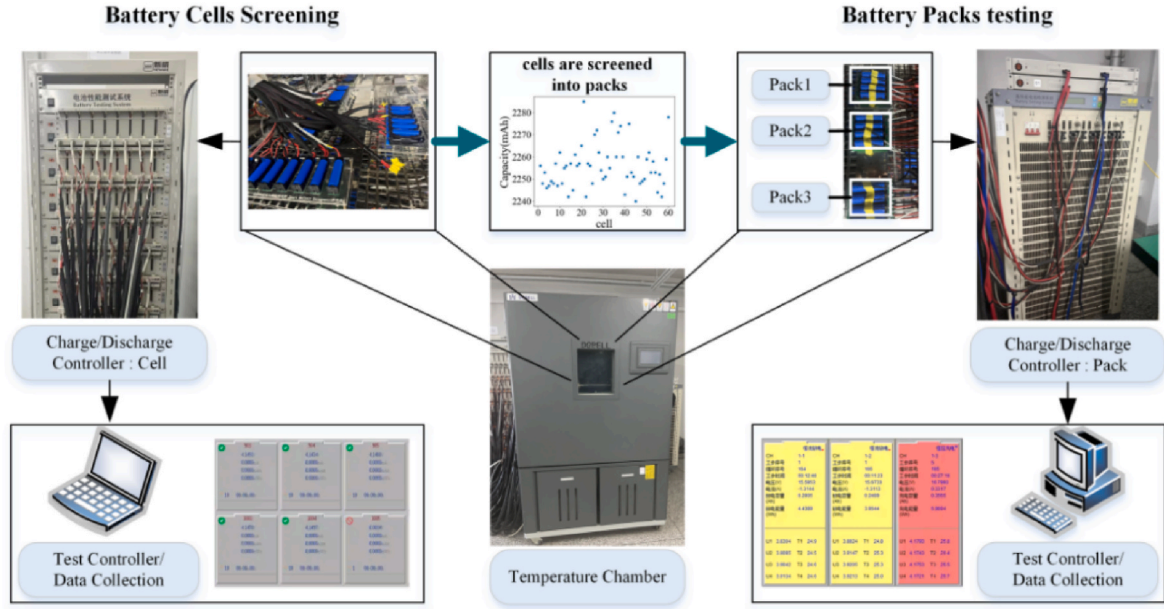


Fig. 1. Test equipment.

above.

3. Proposed method

Addressing battery pack inconsistency representation and degradation modeling, this paper proposes an inconsistency-based degradation modeling method for serial battery packs. The specific framework of the methodology proposed in this paper is shown in Fig. 3.

In Fig. 3, σ_{time} represents the matrix of dynamic inconsistency representation parameters; $\sigma_{time}^i(k)$ denotes the dynamic inconsistency parameters for the k th cycle and the i th moment; m is the total number of cycles, and n refers to the total number of moments. σ_{DOD} is the matrix of static inconsistency representation parameters; $\sigma_{DOD}(k)$ represents the static inconsistency parameter for the k th cycle. σ is the cluster of acquired inconsistency representation parameters. The above parameters are described in detail in subsequent sections. As depicted in Fig. 3, the methodology proposed in this paper consists of two main parts. The first part proposes a method to obtain inconsistency representation parameters, addressing the issue of insufficient online representation capability for battery pack inconsistency. The online-monitored voltage is used to construct both static and dynamic inconsistency representation parameters, enabling the multiple dimensions and multiple moments online representation of battery pack inconsistency, thus enriching the methods of inconsistency representation. The second part proposes an inconsistency-based degradation modeling method to tackle the challenges of significant modeling errors and insufficient nonlinear fitting capability in battery pack degradation models. Linear and nonlinear models are established and subsequently fused. This achieves nonlinear degradation path fitting for the battery pack while enhancing model accuracy.

3.1. Inconsistency representation cluster construction

In the process of inconsistency representation parameter construction, in this paper, serialized dynamic inconsistency representation parameters are constructed from multiple moments using online monitoring voltage with Gaussian distribution. Then, the static inconsistency representation parameters are constructed online by indirectly characterizing the battery DOD and quantifying the DOD differences. Finally, multiple moments, multiple dimensions inconsistency

representation cluster is constructed by joining the two. The details of the above steps are as follows:

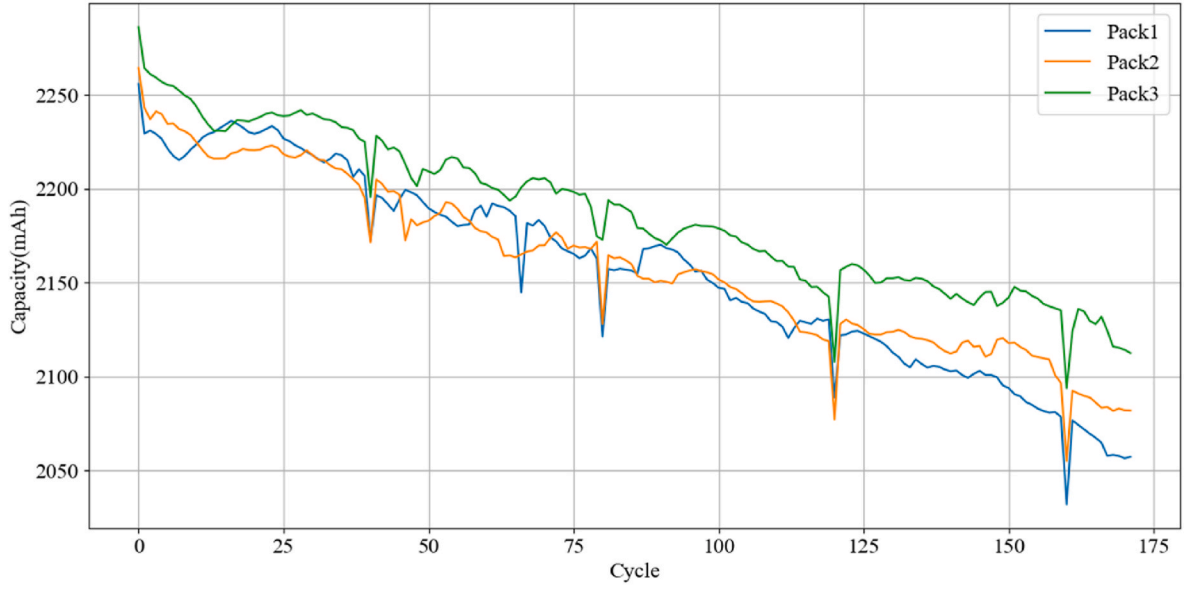
The voltage differences between individual cells in a lithium-ion battery pack fluctuate rapidly at any given moment, forming the basis for deriving dynamic inconsistency representation parameters. Previous studies indicate that parameters such as voltage, capacity, and internal resistance in a lithium-ion battery pack follow a normal distribution [35]. Consequently, Eq. (1) represents the fitting approach utilized in this paper, employing the normal distribution to model each parameter within the battery pack.

$$f(v) = \frac{1}{\sqrt{2\pi}\sigma_{in}} \exp\left(-\frac{(v_{in} - \mu_{in})^2}{2\sigma_{in}^2}\right) \quad (1)$$

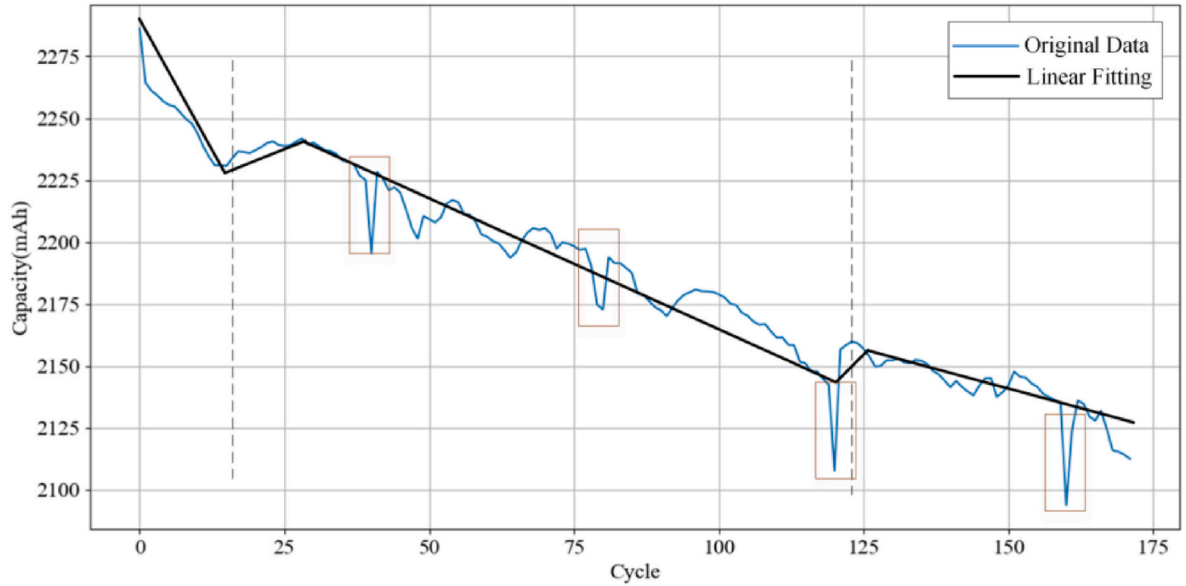
where v_{in} represents the voltage; μ_{in} denotes the mean value of the voltage parameter; and σ_{in} indicates the variance of the voltage parameter. This equation serves as the foundation for representing inconsistency within the battery pack. In this study, the variance is employed to quantify inconsistency within the battery pack. However, the Anderson-Darling test (AD test) must be conducted prior to fitting to confirm that the selected data adheres to the normal distribution. The calculation process for the AD test is outlined in Eq. (2).

$$A^2 = -k - \sum_{i=1}^l \left[\frac{2i-1}{l} \cdot \ln(F(v_{in,i})) + \ln(1 - F(v_{in,l-i+1})) \right] \quad (2)$$

where $F()$ represents the cumulative distribution function and l denotes the total number of cells. Following the normal distribution test, a normal distribution fitting is performed. Using Eq. (1), the voltage data distribution at each moment of the lithium-ion battery pack is fitted to obtain the variance. The voltage variance within the lithium-ion battery pack fluctuates over time. Relying solely on a single moment of voltage data for inconsistency representation compromises the ability to capture the dynamic evolution process. Moreover, the representation parameters are susceptible to noise and other factors, leading to inadequate representation ability. To address these issues, this paper employs the multiple moments voltage variance (referred to as Time-Vol-Var) to construct dynamic inconsistency representation parameter clusters. The structure of the representation parameter cluster is outlined in Eq. (3).



(a) Comparison of battery pack capacity degradation data



(b) Data analysis battery pack3

Fig. 2. Battery pack capacity degradation data.

$$\sigma_{time} = [\sigma_{time}^i(k)]_{m \times n} = \begin{bmatrix} \sigma_{time}^1(1), & \sigma_{time}^2(1), & \dots & \sigma_{time}^n(1) \\ \sigma_{time}^1(2), & \sigma_{time}^2(2), & \dots & \sigma_{time}^n(2) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{time}^1(m), & \sigma_{time}^2(m), & \dots & \sigma_{time}^n(m) \end{bmatrix}_{m \times n} \quad (3)$$

where n denotes the number of moments used, and m denotes the number of cycle periods; $\sigma_{time}^i(k)$ denotes the voltage variance corresponding to the k th cycle and the i th moment. The serialized representation of dynamic inconsistency parameters within serial space lithium-ion battery packs is established by Eq. (3). A more comprehensive representation of the dynamic evolution of voltage within the

battery pack is obtained by utilizing additional time points. Simultaneously, the impact of errors, such as noise, at a single moment is minimized. This, in turn, enhances the representation of dynamic inconsistency in serial space lithium-ion battery packs.

Variations in the DOD among cells in a serial battery pack are the primary cause of differential cell degradation. Moreover, DOD variations also serve as a key indicator of the differences in cell degradation [36]. The slow variation in DOD makes it suitable for constructing static inconsistency representation parameters. In this paper, DOD is characterized by the difference between the start and stop voltages during the discharge of individual cells. Based on Eq. (1), a distribution is fitted to the starting and stopping voltage difference data of the discharge process to obtain the variance in the distribution (referred to as

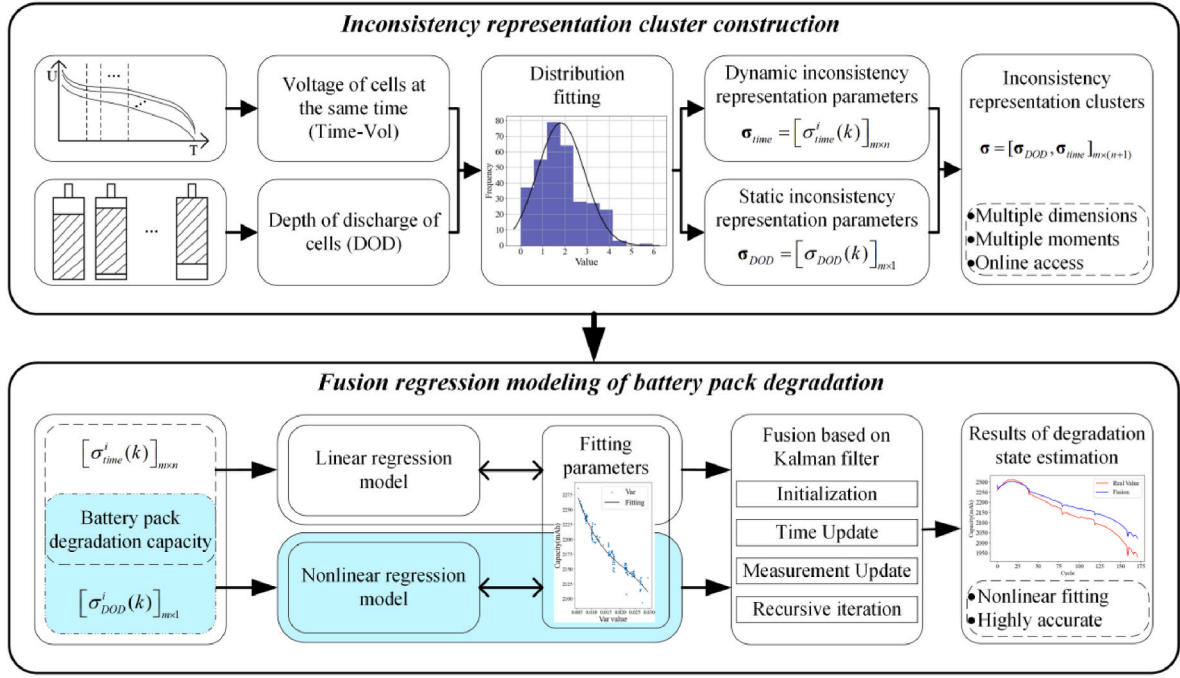


Fig. 3. Block diagram of the overall program.

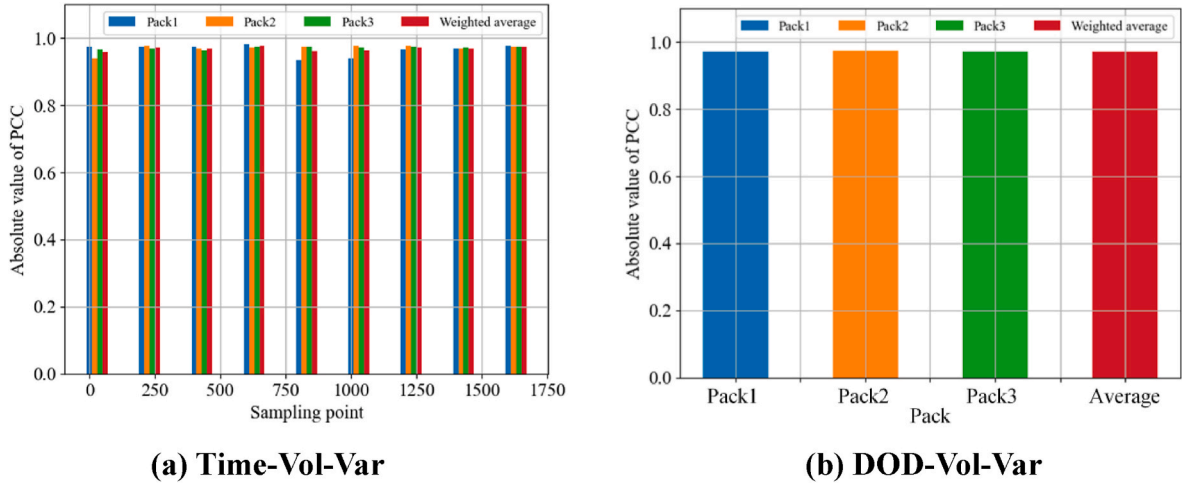


Fig. 4. Inconsistency representation parameter correlation coefficients.

Table 2

Correlation coefficient calculation results.

Parameter	Sampling point	Pack1	Pack2	Pack3	Weighted average
Time-Vol-Var	1	0.9732	0.9401	0.9653	0.9595
	200	0.9731	0.9763	0.9679	0.9724
	400	0.9750	0.9684	0.9640	0.9691
	600	0.9815	0.9727	0.9736	0.9760
	800	0.9347	0.9742	0.9732	0.9607
	1000	0.9411	0.9760	0.9726	0.9633
	1200	0.9658	0.9759	0.9741	0.9720
	1400	0.9688	0.9697	0.9708	0.9698
DOD-Vol-Var	1600	0.9769	0.9746	0.9733	0.9749
	\	0.9706	0.9746	0.9726	0.9726

DOD-Vol-Var). The extracted variance data is expressed as shown in Eq. (4).

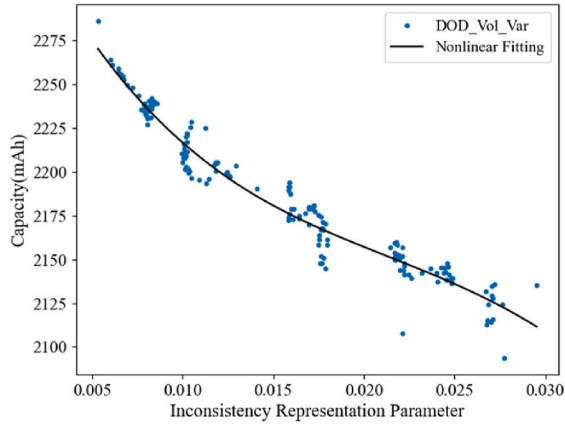
$$\sigma_{DOD} = [\sigma_{DOD}(k)]_{m \times 1} = \begin{bmatrix} \sigma_{DOD}(1) \\ \sigma_{DOD}(2) \\ \sigma_{DOD}(m) \end{bmatrix}_{m \times 1} \quad (4)$$

Using the static inconsistency representation parameters obtained from Eq. (4), the issue of parameters such as capacity and internal resistance not being obtainable online is effectively addressed, enabling the online representation of static inconsistency in serial space lithium-ion battery packs.

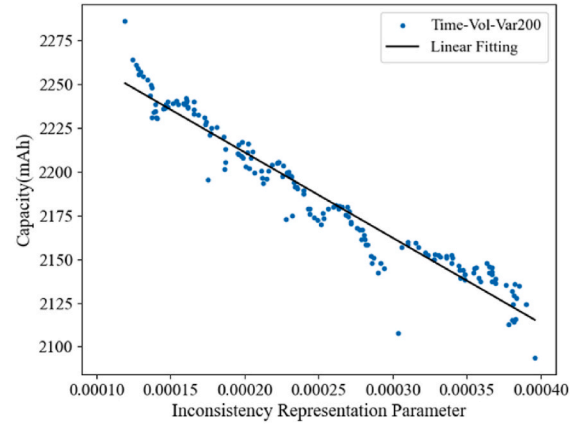
The previously described differences in the battery pack can be consolidated into a cluster that represents inconsistency across multiple dimensions, as demonstrated in Eq. (5).

$$\sigma = [\sigma_{DOD}, \sigma_{time}]_{m \times (n+1)} \quad (5)$$

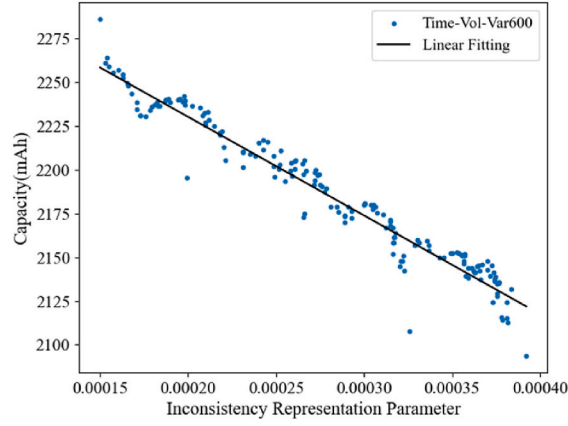
In summary, this paper introduces a novel approach to constructing



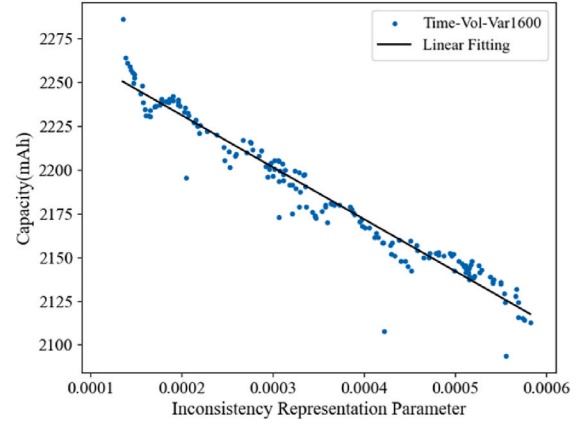
(a) DOD-Vol-Var



(b) Time-Vol-Var200



(c) Time-Vol-Var600



(d) Time-Vol-Var1600

Fig. 5. Trends of degradation curves and inconsistency representation parameter curves.

Table 3

Comparison of linear, nonlinear, and fusion model results.

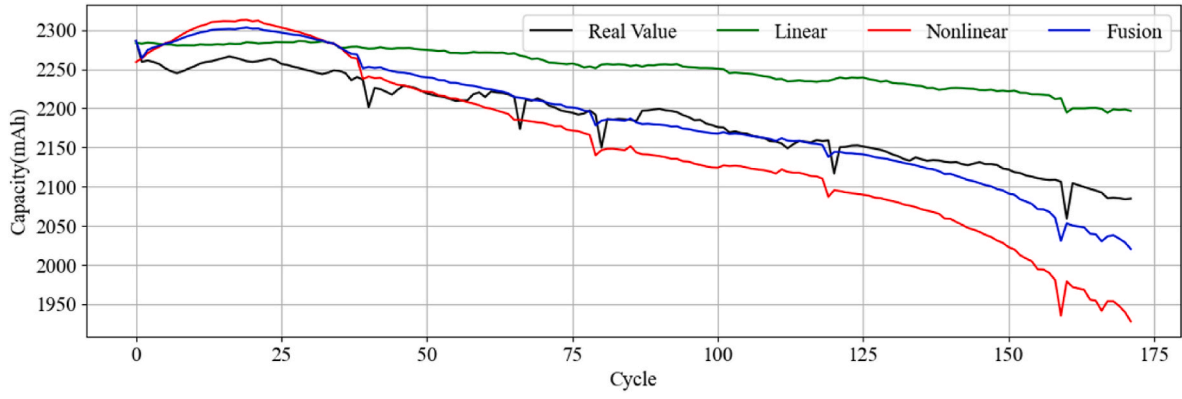
Method	Training dataset Pack1				Training dataset Pack2				Training dataset Pack3			
	Pack2		Pack3		Pack1		Pack3		Pack1		Pack2	
	MAE	ME	MAE	ME	MAE	ME	MAE	ME	MAE	ME	MAE	ME
Linear	5.66 %	9.44 %	3.57 %	5.59 %	3.88 %	7.587 %	2.12 %	4.71 %	2.91 %	5.92 %	0.58 %	3.14 %
Nonlinear	2.05 %	3.73 %	1.47 %	3.63 %	0.67 %	2.55 %	0.64 %	2.78 %	2.26 %	7.48 %	2.04 %	5.44 %
Fusion	1.73 %	3.20 %	1.28 %	2.85 %	0.55 %	1.95 %	0.42 %	1.88 %	0.95 %	3.31 %	0.53 %	3.02 %

inconsistency representation parameters that effectively captures both dynamic and static inconsistencies in lithium-ion battery packs. By utilizing online monitoring voltage data to establish serialized dynamic inconsistency parameters over multiple moments, it overcomes the limitations of single moment representation. At the same time, by using the online monitoring voltage to indirectly characterize the DOD of the battery and quantify the difference of the DOD, the online representation of static inconsistency is realized, which overcomes the problem that the traditional static inconsistency representation parameters, such as capacity, cannot be obtained online. The fusion of these dynamic and static parameters forms a multiple dimensions and multiple moments inconsistency representation cluster, offering a more comprehensive and

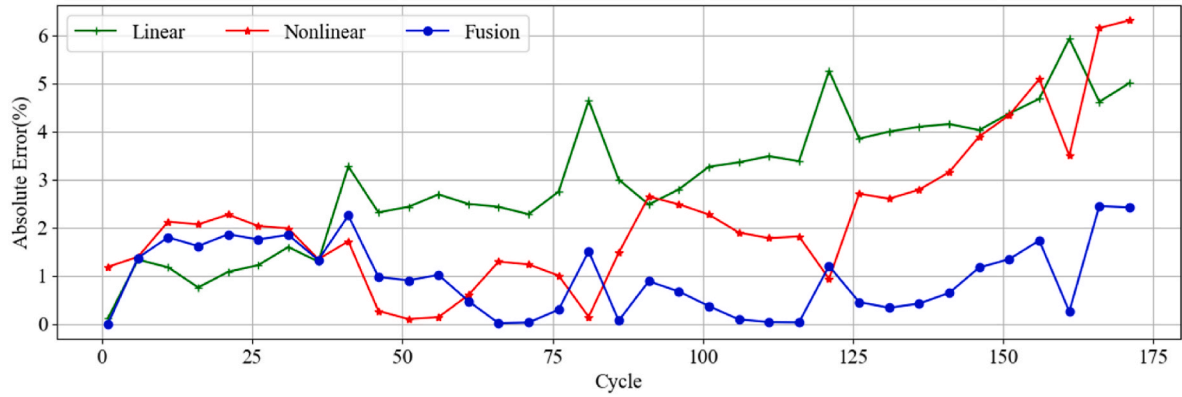
innovative method for accurately representing and analyzing battery pack inconsistencies.

3.2. Fusion regression modeling of battery pack degradation

In the process of degradation modeling, this paper utilizes the dynamic inconsistency representation parameter for linear regression model construction to achieve multi-source information fitting. Then, a nonlinear regression model is constructed using static inconsistency representation parameters to capture the nonlinear degradation trend of the battery pack. Finally, the two models are fused using Kalman filter to improve the modeling accuracy. The specific steps of the above process



(a) Comparison of results before and after fusion



(b) Comparison of errors before and after fusion

Fig. 6. Comparison of before and after fusion.

are as follows:

The previous section identified two dimensions of battery pack inconsistency parameters: static and dynamic. Dynamic inconsistency representation parameters offer richer informational content due to their multiple dimensions temporal characteristics and significant disparities. Moreover, the multiple dimensions linear regression model is relatively straightforward. This model effectively mitigates the risks of over-complexity, parameter identification difficulties, and overfitting. Thus, this research employs the dynamic inconsistency parameters to develop a multiple dimensions linear regression model, as represented by Eq. (6).

$$C_{est}(k) = \omega_1 * \sigma_{time}^1(k) + \omega_2 * \sigma_{time}^2(k) + \dots + \omega_n * \sigma_{time}^n(k) + \omega_0 \quad (6)$$

where ω_i denotes the parameter weight corresponding to the i th inconsistency parameter; ω_0 denotes the bias; and C_{est} denotes the battery pack performance degradation state. The above weighting parameters and biases are the parameters to be identified. Assuming that ω_0 corresponds to $\sigma_{time}^0(k)$ and that ω_0 is always 1, each parameter to be identified can be represented as Eq. (7).

$$\omega = [\omega_0, \omega_1, \dots, \omega_n]_{1 \times (n+1)} \quad (7)$$

Further, Eq. (6) is equated to Eq. (8).

$$C_{est}(k) = \sigma_{time}^T(k) \times \omega \quad (8)$$

The parameters involved in Eq. (7) are next estimation. The traditional least squares method is susceptible to 'data explosion' during parameter estimation, which compromises the accuracy of parameter acquisition. To address this, the least squares forgetting factor method

incorporates a forgetting mechanism to prevent data explosion. Hence, this method is chosen for parameter estimation in this paper. The detailed process is as follows:

Obtain the parameter estimation in Eq. (8) based on Eq. (9).

$$\begin{cases} \omega(k) = \omega(k-1) + K(k)(C_{true}(k) - C_{est}(k)) \\ K(k) = P(k-1)\sigma_{time}(k)[\sigma_{time}^T(k)P(k-1)\sigma_{time}(k) + \lambda]^{-1} \\ P(k) = P(k-1)[\sigma_{time}^T(k)P(k-1)\sigma_{time}(k+1) + \lambda]^{-1} \end{cases} \quad (9)$$

where $C_{true}(k)$ denotes the true value of battery pack performance degradation at cycle k ; $P(k)$ is the covariance matrix; $K(k)$ is the gain matrix; and λ is the forgetting factor.

The nonlinear characteristics of lithium-ion batteries during degradation must be fitted with a nonlinear model. Relying solely on a linear model (e.g., Eq. (8)) results in estimation errors. Thus, this paper employs a nonlinear regression model to fit the battery degradation process, capturing battery performance degradation behavior more accurately. Considering the complexity of multiple dimensions nonlinear regression models and the risk of overfitting, this paper constructs a one-dimension nonlinear regression model using the static inconsistency representation parameter; the model's specific form is shown in Eq. (10). This approach not only simplifies the model structure but also enhances the accuracy of fitting the battery degradation process.

$$C_{Non}(k) = \beta_3 \exp\left(-\frac{(\sigma_{DOD}(k) - \beta_2)^2}{\beta_1}\right) + \beta_6 \exp\left(-\frac{(\sigma_{DOD}(k) - \beta_5)^2}{\beta_4}\right) \quad (10)$$

Table 4
Comparison results of different methods.

Training data	Pack1		Pack3	
Testing data	Pack2		Pack3	
Indicators	MAE	ME	MAE	ME
Fusion model	1.73 %	3.20 %	1.28 %	2.85 %
KF-RVM	2.33 %	3.51 %	1.47 %	3.12 %
GPR	2.82 %	4.61 %	4.30 %	5.63 %
SVR	2.43 %	4.55 %	1.55 %	4.08 %
TL-LSTM	1.74 %	5.27 %	1.57 %	4.67 %
FNN	1.77 %	4.03 %	1.79 %	4.96 %

Training data	Pack2		Pack3	
Testing data	Pack1		Pack2	
Indicators	MAE	ME	MAE	ME
Fusion model	0.55 %	1.95 %	0.42 %	1.88 %
KF-RVM	0.97 %	2.69 %	0.75 %	2.78 %
GPR	1.55 %	5.36 %	1.05 %	2.53 %
SVR	1.61 %	5.09 %	1.13 %	3.47 %
TL-LSTM	2.29 %	7.22 %	1.72 %	5.05 %
FNN	3.10 %	6.96 %	2.32 %	6.53 %

Training data	Pack3		Pack2	
Testing data	Pack1		Pack2	
Indicators	MAE	ME	MAE	ME
Fusion model	0.95 %	3.31 %	0.53 %	3.02 %
KF-RVM	1.04 %	3.56 %	0.89 %	3.38 %
GPR	1.57 %	4.14 %	0.85 %	3.30 %
SVR	1.79 %	3.76 %	1.91 %	3.85 %
TL-LSTM	2.27 %	7.17 %	3.12 %	6.64 %
FNN	1.98 %	5.37 %	1.54 %	4.24 %

where $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5$, and β_6 denote the individual parameter, respectively. The parameters above can be estimated using the least squares forgetting factor method. The construction of a linear model and a nonlinear model is completed, and the two models have the advantages of multi-source information fitting and nonlinear fitting, respectively, and the fusion of the two models can further improve the accuracy of the model. Based on this, Eq. (6) serves as the state transition equation, and Eq. (10) is the observation equation. Model fusion is performed using the Kalman filter to achieve model complementarity. The final model is represented in Eq. (11).

$$\begin{cases} C_{sta}(k) = C_{est}(k) + \theta_1 \\ C_{oba}(k) = C_{Non}(k) + \theta_2 \end{cases} \quad (11)$$

where $C_{sta}(k)$ is the state transfer value of the k th cycle; $C_{oba}(k)$ is the observation value of the k th cycle; θ_1 and θ_2 are the state transfer noise and observation noise, respectively. The optimal state estimate $\widehat{C_{est}}(k)$ is obtained by fusion model.

In summary, this paper introduces a novel approach to degradation modeling by integrating both linear and nonlinear regression techniques, addressing key limitations in existing methods. Specifically, the dynamic inconsistency representation parameters are employed for multi-source information fitting through linear regression, while static inconsistency parameters are captured through a nonlinear model to account for the complex degradation behavior of battery packs. The fusion of these models using a Kalman filter further enhances the overall accuracy. This fusion framework effectively leverages the complementary strengths of linear and nonlinear modeling, resulting in a more precise and adaptable degradation estimation.

4. Experimental results and discussion

4.1. Experimental design

A total of three experiments are conducted to prove the effectiveness

of the method proposed in this paper.

- (1) Validation of the validity of inconsistency parameters of lithium-ion battery packs

Correlation analyses are conducted between the battery pack inconsistency parameters and the maximum discharge capacity. This correlation illustrates the representational capacity of the quantitative parameters, indicating their suitability for modeling.

- (2) Validation of ablation experiments of the fusion modeling approach proposed

The linear model, nonlinear model, and fusion model proposed in this paper are evaluated separately using multiple test datasets. Comparative analyses demonstrate that the performance of the fusion model surpasses that of the individual models, thereby validating the methodology proposed in this paper.

- (3) Validation of the validity of the fusion modeling approach proposed

More common lithium-ion battery degradation modeling methods are selected from references for comparison, including the method of fusion of Kalman filter and relevance vector machine (KF-RVM) [37]; Gaussian process regression method (GPR) [38]; support vector regression method (SVR) [39]; transfer learning method based on the long and short-term memory network, which in this paper utilizes the test data of 100 cycles as the fine-tuning data (TL-LSTM) [40]; feed-forward neural network method (FNN) [41].

4.2. Evaluation indicators

- (1) Pearson correlation coefficient (PCC)

Pearson correlation coefficient is mainly used to evaluate the correlation between two sequences, and this evaluation index is used to evaluate the first experiment. The calculation process is shown in Eq. (12).

$$PCC = \frac{COV(D_{cmp}, C)}{\sigma_{D_{cmp}} \sigma_C} \quad (12)$$

where D_{cmp} denotes the sequence to be compared, which in this paper refers to each inconsistency parameter; $\sigma_{D_{cmp}}$ denotes the variance of the sequence to be compared; and σ_C denotes the variance of the degraded state of the battery pack performance.

- (2) Mean absolute error (MAE)

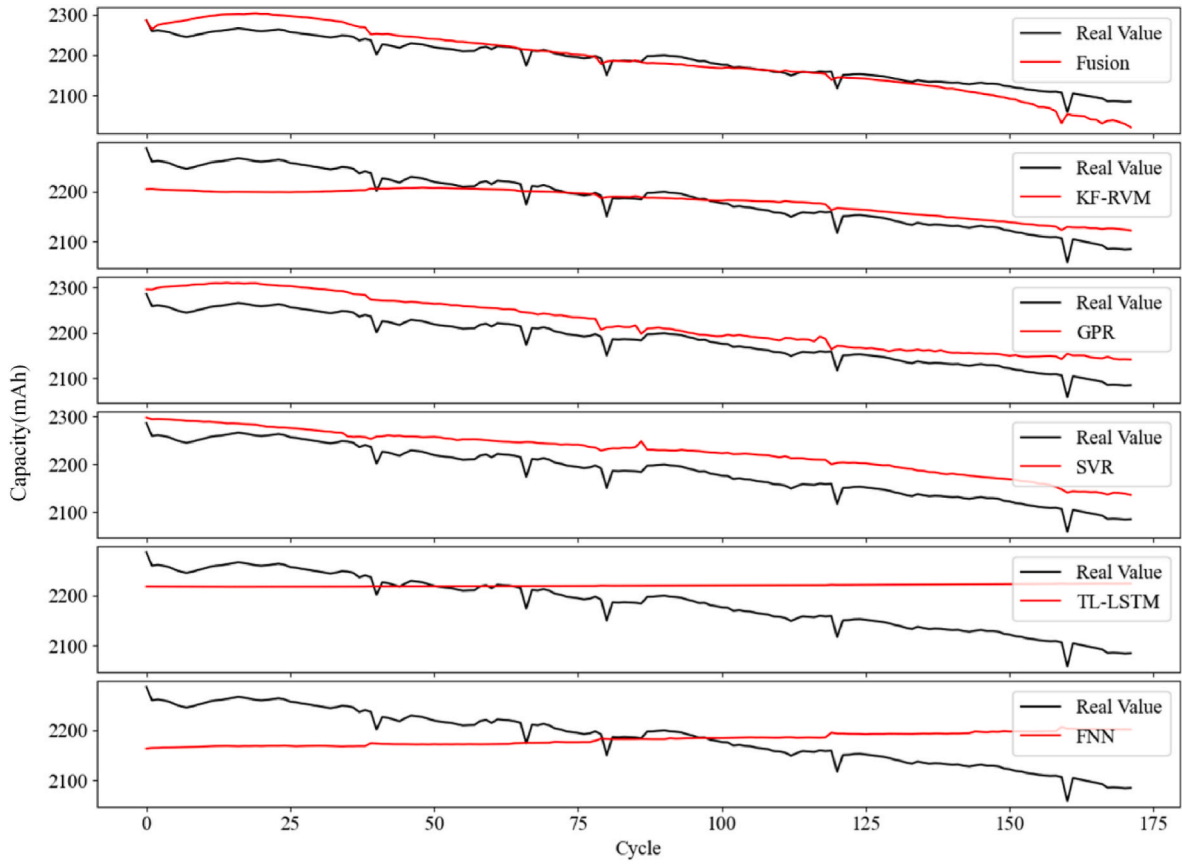
MAE can evaluate the performance of the method globally for evaluating the second experiment and the third experiment. The calculation process is shown in Eq. (13).

$$MAE = \frac{1}{m} \sum_{k=1}^m |\widehat{C_{est}}(k) - C_{true}(k)| \quad (13)$$

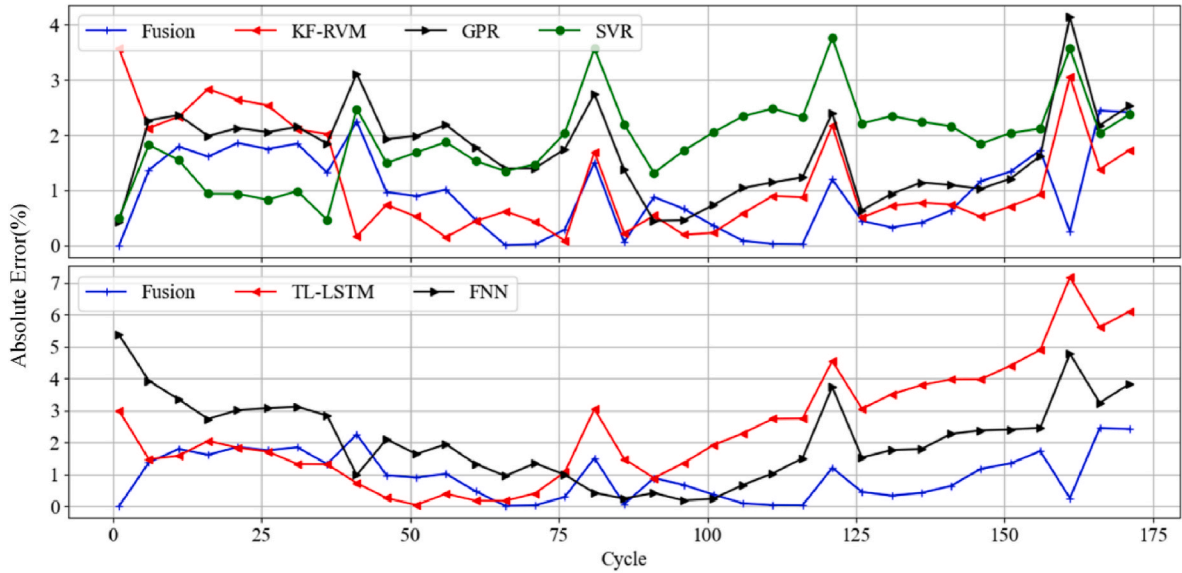
- (3) Maximum error (ME)

ME can evaluate the performance of the method locally for the evaluation of the second experiment with the third experiment. The calculation process is shown in Eq. (14).

$$ME = \max \{ \widehat{C_{est}}(k) - C_{true}(k) \}_{k=1}^m \quad (14)$$



(a) Comparison of results of different methods



(b) Comparison of errors of different methods

Fig. 7. Comparison of different methods.

4.3. Validation of the validity of inconsistency representation parameters

With an increasing number of charge and discharge cycles, the discharge time of the lithium-ion battery decreases. To ensure simultaneous sampling across all cycles, the maximum sampling point for the

dynamic inconsistency representation parameter (Time-Vol-Var) is determined by the shortest discharge duration. The sampling process is limited to a maximum of 1600 points, with computations conducted at intervals of 200 points, beginning from 1. The variance of the start and stop voltage difference data (DOD-Vol-Var) during the discharge process

serves as the static inconsistency representation parameter. The aforementioned data sets were tested for normal distribution and were therefore suitable for fitting to a normal distribution. The PCC between the obtained variance of each inconsistency representation parameter and the maximum discharge capacity was calculated, with the corresponding results presented in Fig. 4 and Table 2.

Generally, if the absolute value of the PCC exceeds 0.80, it indicates a strong correlation between the parameters, suggesting they can effectively represent each other. As shown in Fig. 4 and Table 2, the absolute values of the correlation coefficients for the parameters are all greater than 0.90, confirming their strong representational ability. However, the representation ability of certain parameters fluctuates, with more pronounced variations observed in the Time-Vol-Var parameter. To ensure consistency among the model's input parameters, weights are assigned to the three datasets, which are then averaged to finalize the representation parameters. In Table 2, parameters with correlation coefficients exceeding 0.80 were selected using the forward variable selection method. Based on the highest correlation coefficient, parameters were tested sequentially. Finally, the 200, 600, and 1600 moment parameters were selected as the dynamic inconsistency representation parameters. To further illustrate the battery pack's inconsistency representation parameters, these selected parameters are plotted in Fig. 5. Due to space constraints, only the results for battery pack 3 are presented in Fig. 5.

Fig. 5(a) illustrates that the DOD-Vol-Var exhibits a strong nonlinear fitting relationship with degradation. Fig. 5(b), (c), and 5(d) exhibits that the Time-Vol-Var demonstrates a robust linear fitting relationship with degradation. This further validates the effectiveness of using these parameters for both nonlinear and linear adjustments. The differences among Fig. 5(b), (c), and 5(d) are minimal, attributed to the short sampling intervals. The effectiveness of the inconsistency representation parameters constructed using online monitorable parameters in this paper is further demonstrated. It effectively compensates the problem that static inconsistency representation parameters such as capacity cannot be obtained online, and realizes the serialized expression of dynamic inconsistency representation parameters. The online representation of multiple dimensions and multiple moments inconsistency of serial lithium-ion battery pack can be realized by the above parameters. In summary, the extracted parameters, as shown in both Table 2 and Fig. 5, exhibit a strong correlation with the degraded state of the battery, providing a solid foundation for subsequent modeling efforts.

4.4. Validation of ablation experiments of the fusion modeling approach proposed

To validate the fusion model proposed in this paper, the results of the linear, nonlinear, and fusion models are compared and validated. The three sets of battery pack data are used sequentially as the training set and test set. The results obtained from the experiments are shown in Table 3. Due to space constraints, this paper only plots the test results and errors when battery pack 3 is used as the training set and battery pack 1 is used as the test set, as shown in Fig. 6.

As shown in Table 3, the proposed fusion method exhibits the smallest error compared to the other methods. When battery pack 3 is used as the training set and battery pack 1 as the test set, the linear method demonstrates lower error than the nonlinear method. Furthermore, the linear methods consistently show the largest errors. This can be attributed to the simplicity of the linear model structure, where even minor changes in the input data can significantly affect the output, resulting in larger errors. Despite having a greater error, the linear model remains essential. As shown in Fig. 6, the estimation error of the linear model is lower than that of the nonlinear model at both the beginning and end of battery degradation; however, in the middle of degradation, the nonlinear model exhibits a lower error. This phenomenon is attributed to the higher fit of nonlinear models to the data. Greater volatility in the early stages of degradation, as well as the

accumulation of long-term errors in the later stages, may lead to larger estimation errors in the nonlinear model. Thus, this confirms the possibility of complementary strengths between the two models. The fusion model achieved error minimization in the middle of degradation, although the error is larger at the beginning due to the influence of the nonlinear model. Overall, the fusion model proposed in this paper demonstrates superior performance, validating the effectiveness of the methodology presented. It is verified that the method proposed in this paper demonstrates better stability and estimation accuracy by integrating a linear model constructed based on multiple information sources with a nonlinear model with nonlinear fitting capability.

4.5. Validation of the validity of the fusion modeling approach proposed

To further demonstrate the effectiveness of the methods proposed in this paper, five common methods are selected for comparative testing. Three sets of battery packs are used as the training set respectively, and the other two sets of battery data are used as the test set for validation, and the results obtained from the experiments are shown in Table 4. Due to space constraints, this paper only plots the test results and errors when battery pack 3 is used as the training set and battery pack 1 is used as the test set, as shown in Fig. 7 below.

Table 4 shows that the fusion method proposed here exhibits the lowest error among all experiments. Fig. 7 is used to analyze the proposed method and the comparative experimental results in detail. As shown in Fig. 7, due to the small number of samples involved in model training, TL-LSTM and FNN both methods that require a high number of data samples exhibited convergence with significant errors. Therefore, these methods will not be analyzed further. In the detailed comparative analysis of the fusion method, KF-RVM, GPR, and SVR methods in Fig. 7 (a) and (b), the following features can be observed: (1) All methods show large errors in the early stage of battery degradation, likely due to the complexity of performance changes during initial use, making it difficult for the model to capture these changes accurately. In this stage, the proposed method demonstrates the second smallest error and exhibits better adaptability and estimation accuracy. (2) In the middle stage of battery degradation, the proposed method achieves the smallest error, indicating its effectiveness in reducing estimation error and providing high accuracy during periods of more stable battery performance changes. (3) As the battery enters the late stage of degradation, errors for all methods increase, with prolonged estimation leading to error accumulation. Although the error for the proposed method increases, it remains very close to the errors of other methods, demonstrating good stability. In summary, while the proposed method may not consistently minimize error globally, its overall performance is the most stable, with the lowest total error and a maximum error maintained at a relatively low level. This suggests that the method may be more reliable in practical applications. The significant error primarily occurs around the 40th and 80th cycles, as this paper disassembles the experimental battery packs and tests the battery cells every 40 cycles, resulting in considerable data fluctuations after reassembly. The above experimental results further demonstrate the effectiveness of the proposed method.

5. Conclusions

To model the correlation between degradation and inconsistency of serial space lithium-ion battery packs, this paper proposes a method to model the degradation of these battery packs using online inconsistency representation parameters. This method enables online representation of static inconsistency in battery packs and sequential online representation of dynamic inconsistency, addressing the limitations in inconsistency representation and enriching the means of online representation. Based on this, linear and nonlinear models are constructed to fit the degradation paths of battery packs, and a fusion strategy is employed to combine the two models. This effectively captures the nonlinear degradation characteristics of the battery packs, while also reducing

modeling error and improving estimation accuracy. The absolute values of the correlations between the inconsistency representation parameters and battery pack degradation are better than 0.9679, with the mean absolute error superior to 1.73 % and the mean error better than 3.31 %, thus validating the effectiveness of the proposed method. Future research will further validate the method's effectiveness in diverse application scenarios and aim to enhance the interpretability of the modeling equation. The methodology proposed in this paper holds significant academic and practical value for characterizing inconsistency and estimating the degradation state of tandem space lithium-ion battery packs. Based on this methodology, future research can further optimize the inconsistency within battery packs, thereby extending their working life and improving overall performance.

CRedit authorship contribution statement

Yuhang Du: Writing – original draft, Methodology, Investigation, Data curation. **Yuchen Song:** Writing – review & editing, Project administration, Funding acquisition. **Datong Liu:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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